

sigrok Pylon BMS Decoder Capture Analysis

Generated from offline sigrok/PulseView analysis outputs

Executive Summary

30

total captures

17

RS485/UART captures

13

CAN captures

19 / 6 / 5

Bridge / Forward / Direct

Key observations

- Anenji Pylon RS485 JKBMS shows the clearest RS485 latency change: Direct cable is -4581.729 (-47.2%) versus Bridge for request-to-response average.
- Growatt RS485 JKBMS has a much longer Direct cable tail latency: Direct cable is +21480.838 (+40.1%) versus Bridge for full-exchange P95.
- Growatt CAN SeplosBMS cycles are shorter in Direct cable mode: Direct cable is -164247.782 (-19.3%) versus Bridge for cycle average.
- Growatt CAN JKBMS frame rate stays effectively flat across modes: Direct cable is +0.100 (+0.2%) versus Bridge.

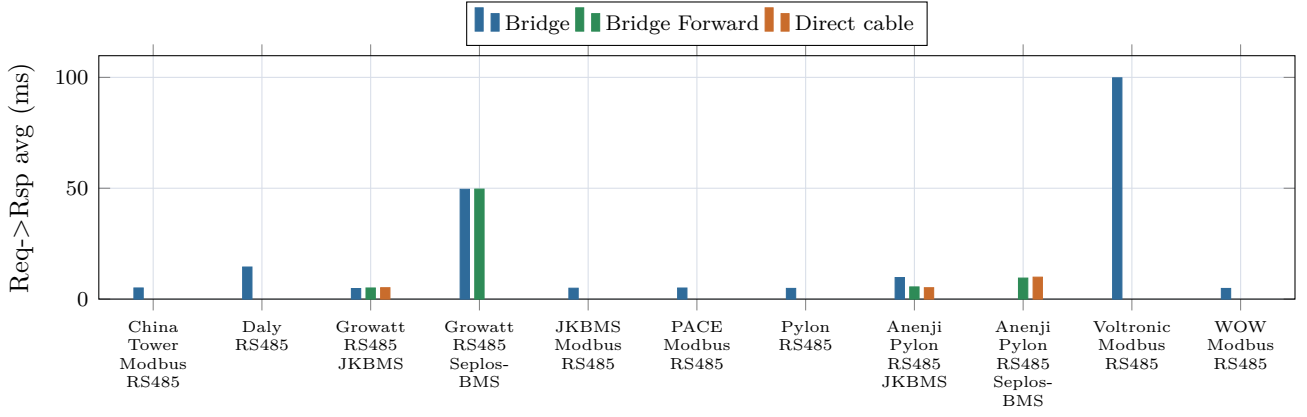
Table 1: Largest Direct cable versus Bridge changes by absolute percentage.

Group	Metric	Direct vs Bridge	Abs. change
Growatt CAN JKBMS	Cycles/s	+0.100 (+100.0%)	100.0%
Growatt CAN JKBMS	Cycle min (us)	-5000467.180 (-50.1%)	50.1%
Growatt CAN JKBMS	Cycle avg (us)	-5000467.180 (-50.1%)	50.1%
Growatt CAN JKBMS	Cycle max (us)	-5000467.180 (-50.1%)	50.1%
Growatt CAN JKBMS	Cycle P95 (us)	-5000467.180 (-50.1%)	50.1%
Growatt CAN SeplosBMS	Cycle min (us)	-121301.900 (-48.3%)	48.3%

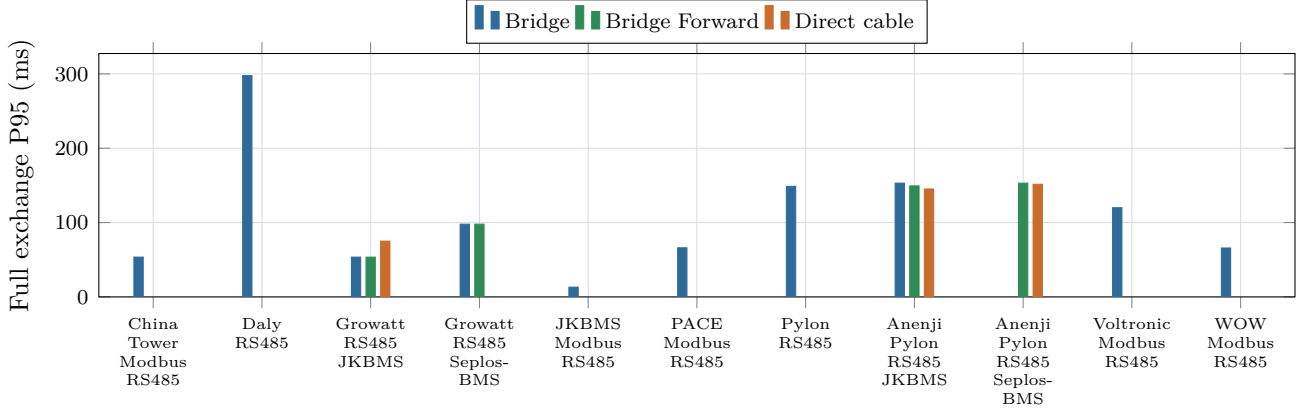
1 Protocol Overview Charts

These charts include every protocol group available in the overview CSV. Missing topology modes are intentionally left blank, so a group can have one, two, or three bars depending on the captures currently available.

CAN cycle duration charts use a logarithmic y-axis because values span multiple orders of magnitude, from sub-millisecond single-frame cycles to multi-second capture-length cycles.

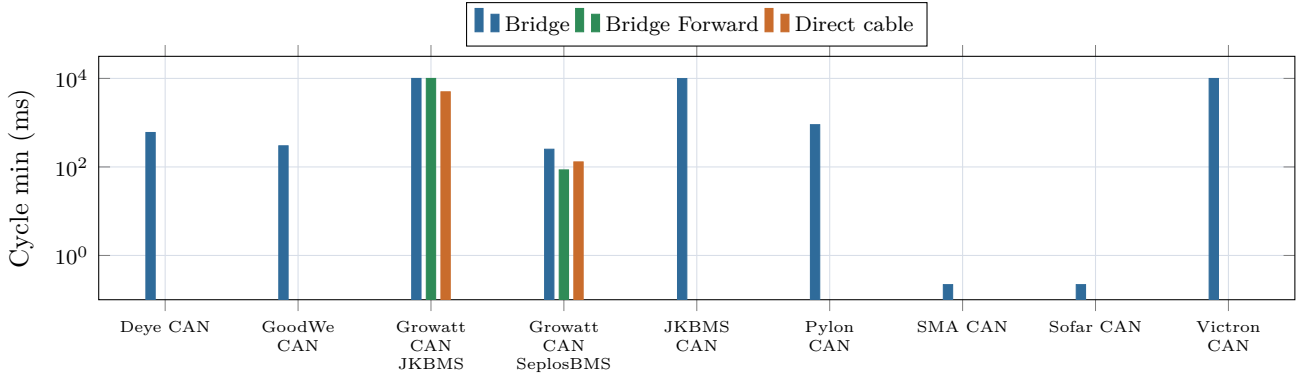


(a) Average request-to-response latency for all available RS485/UART protocol groups.

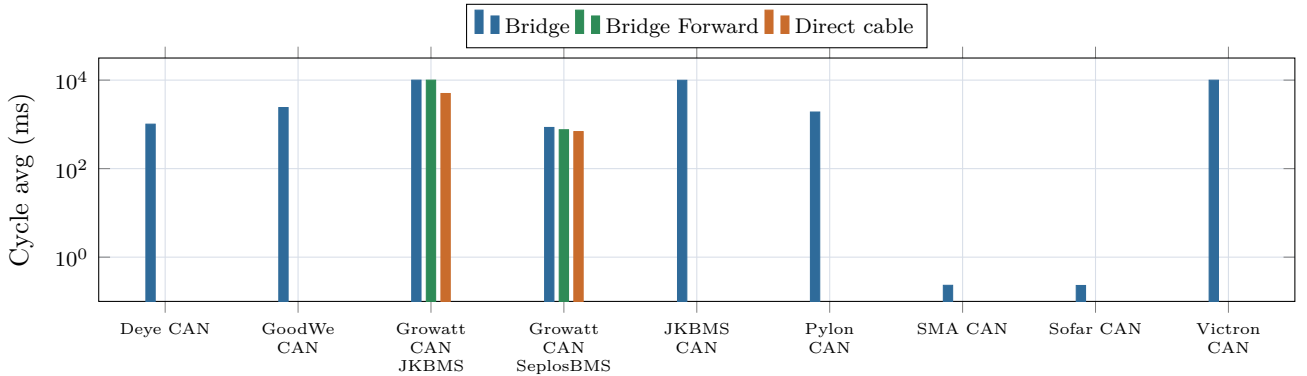


(b) P95 full-exchange duration for all available RS485/UART protocol groups.

Figure 1: RS485/UART protocol overview charts.

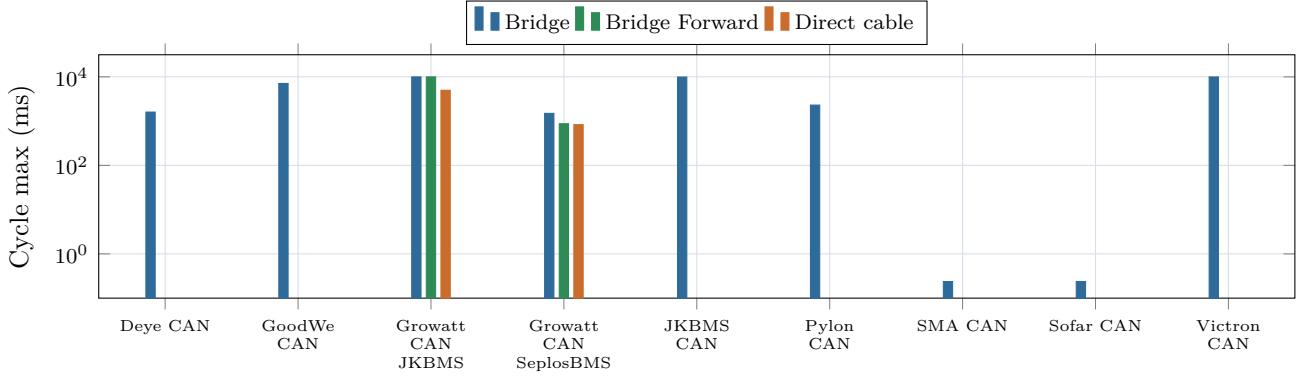


(a) Minimum CAN cycle duration, log scale, for all available CAN protocol groups.

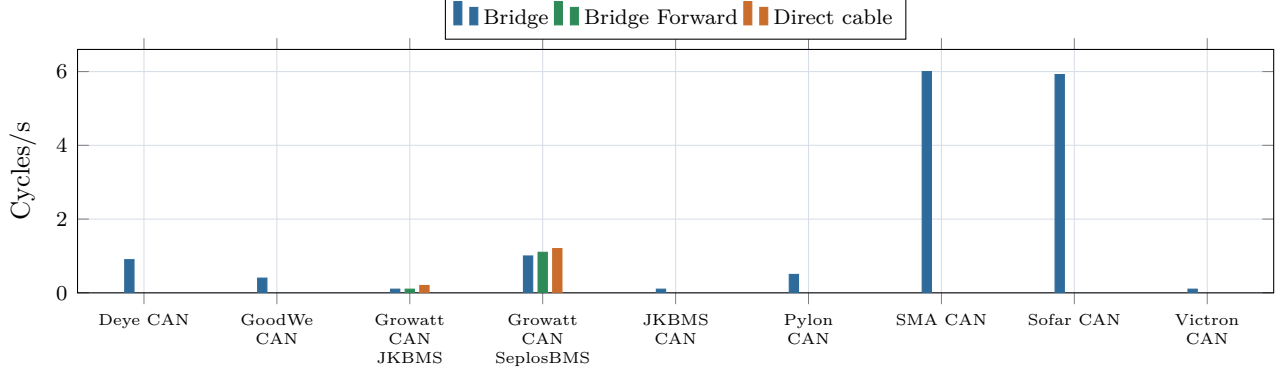


(b) Average CAN cycle duration, log scale, for all available CAN protocol groups.

Figure 2: CAN cycle duration overview charts.

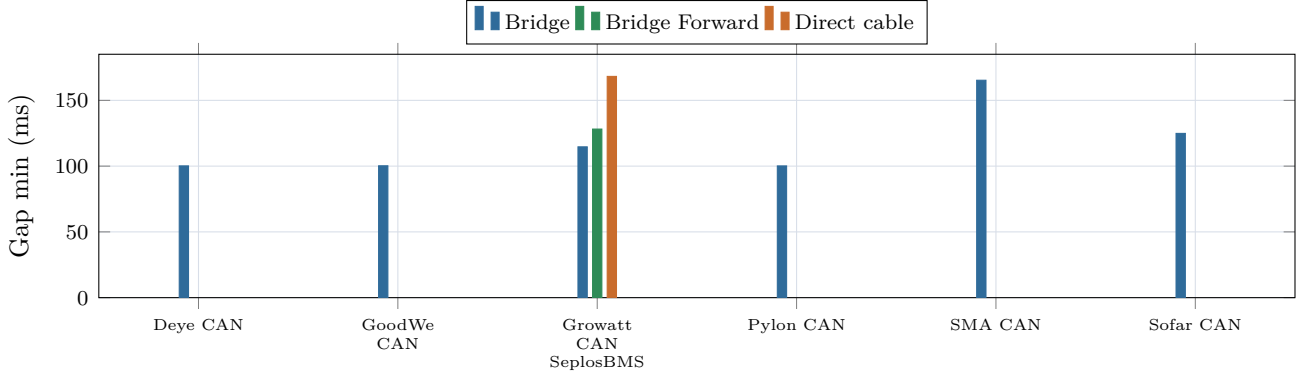


(a) Maximum CAN cycle duration, log scale, for all available CAN protocol groups.

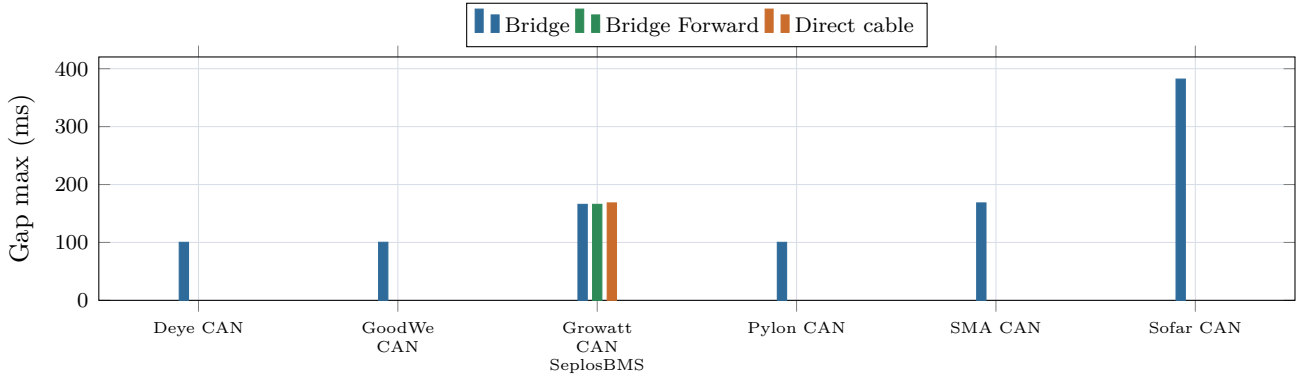


(b) CAN cycle rate normalized by capture duration for all available CAN protocol groups.

Figure 3: CAN maximum cycle duration and throughput charts.



(a) Minimum inter-cycle gap for CAN protocol groups with at least two detected cycles.



(b) Maximum inter-cycle gap for CAN protocol groups with at least two detected cycles.

Figure 4: CAN inter-cycle gap overview charts.

2 Three-mode Delta Tables

The following tables focus only on groups where all three modes are available: Bridge, Bridge Forward, and Direct cable. Deltas are directional: orange means an increase, green means a decrease. Whether an increase is good depends on the metric.

2.1 RS485/UART Three-mode Deltas

Group				Metric	Bridge	Forward	Direct	Forward vs Bridge	Direct vs Bridge	Direct vs Forward
Growatt RS485 JKBMS				Complete exchanges/s	2.500	2.500	2.000	+0.000 (+0.0%)	-0.500 (-20.0%)	-0.500 (-20.0%)
Growatt RS485 JKBMS				Req->Rsp avg (us)	4759.997	5008.099	5126.758	+248.102 (+5.2%)	+366.761 (+7.7%)	+118.659 (+2.4%)
Growatt RS485 JKBMS				Req->Rsp P95 (us)	5294.317	5293.782	5438.647	-0.535 (-0.0%)	+144.330 (+2.7%)	+144.865 (+2.7%)
Growatt RS485 JKBMS				Full exchange avg (us)	49094.793	48926.211	59337.414	-168.582 (-0.3%)	+10242.621 (+20.9%)	+10411.203 (+21.3%)
Growatt RS485 JKBMS				Full exchange P95 (us)	53520.070	53484.774	75000.908	-35.296 (-0.1%)	+21480.838 (+40.1%)	+21516.134 (+40.2%)
Anenji JKBMS	Pylon	RS485		Complete exchanges/s	0.800	0.800	0.800	-0.000 (-0.0%)	-0.000 (-0.0%)	+0.000 (+0.0%)
Anenji JKBMS	Pylon	RS485		Req->Rsp avg (us)	9706.670	5488.619	5124.941	-4218.051 (-43.5%)	-4581.729 (-47.2%)	-363.678 (-6.6%)
Anenji JKBMS	Pylon	RS485		Req->Rsp P95 (us)	9912.793	6190.079	5692.958	-3722.714 (-37.6%)	-4219.835 (-42.6%)	-497.121 (-8.0%)
Anenji JKBMS	Pylon	RS485		Full exchange avg (us)	111164.319	106950.364	102917.894	-4213.955 (-3.8%)	-8246.425 (-7.4%)	-4032.470 (-3.8%)
Anenji JKBMS	Pylon	RS485		Full exchange P95 (us)	153034.215	149320.428	145154.533	-3713.787 (-2.4%)	-7879.682 (-5.1%)	-4165.895 (-2.8%)

2.2 CAN Three-mode Deltas

Group	Metric	Bridge	Forward	Direct	Forward vs Bridge	Direct vs Bridge	Direct vs Forward
Growatt CAN JKBMS	Frames/s	51.900	52.000	52.000	+0.100 (+0.2%)	+0.100 (+0.2%)	+0.000 (+0.0%)
Growatt CAN JKBMS	Cycles/s	0.100	0.100	0.200	+0.000 (+0.0%)	+0.100 (+100.0%)	+0.100 (+100.0%)
Growatt CAN JKBMS	Cycle min (us)	9980699.265	9980680.500	4980232.085	-18.765 (-0.0%)	-5000467.180 (-50.1%)	-5000448.415 (-50.1%)
Growatt CAN JKBMS	Cycle avg (us)	9980699.265	9980680.500	4980232.085	-18.765 (-0.0%)	-5000467.180 (-50.1%)	-5000448.415 (-50.1%)
Growatt CAN JKBMS	Cycle max (us)	9980699.265	9980680.500	4980232.085	-18.765 (-0.0%)	-5000467.180 (-50.1%)	-5000448.415 (-50.1%)
Growatt CAN JKBMS	Cycle P95 (us)	9980699.265	9980680.500	4980232.085	-18.765 (-0.0%)	-5000467.180 (-50.1%)	-5000448.415 (-50.1%)
Growatt CAN Seplos-BMS	Frames/s	13.000	12.900	13.000	-0.100 (-0.8%)	+0.000 (+0.0%)	+0.100 (+0.8%)
Growatt CAN Seplos-BMS	Cycles/s	1.000	1.100	1.200	+0.100 (+10.0%)	+0.200 (+20.0%)	+0.100 (+9.1%)
Growatt CAN Seplos-BMS	Cycle min (us)	251234.865	85691.125	129932.965	-165543.740 (-65.9%)	-121301.900 (-48.3%)	+44241.840 (+51.6%)
Growatt CAN Seplos-BMS	Cycle avg (us)	851373.724	755942.000	687125.942	-95431.724 (-11.2%)	-164247.782 (-19.3%)	-68816.058 (-9.1%)
Growatt CAN Seplos-BMS	Cycle max (us)	1496223.985	871814.905	831769.225	-624409.080 (-41.7%)	-664454.760 (-44.4%)	-40045.680 (-4.6%)
Growatt CAN Seplos-BMS	Cycle P95 (us)	1221318.481	861952.157	831768.787	-359366.324 (-29.4%)	-389549.694 (-31.9%)	-30183.370 (-3.5%)
Growatt CAN Seplos-BMS	Inter-cycle gap min (us)	114668.755	128174.720	168225.685	+13505.965 (+11.8%)	+53556.930 (+46.7%)	+40050.965 (+31.2%)
Growatt CAN Seplos-BMS	Inter-cycle gap avg (us)	155493.468	160226.810	168228.483	+4733.342 (+3.0%)	+12735.015 (+8.2%)	+8001.673 (+5.0%)
Growatt CAN Seplos-BMS	Inter-cycle gap max (us)	165777.835	165777.835	168229.680	+0.000 (+0.0%)	+2451.845 (+1.5%)	+2451.845 (+1.5%)
Growatt CAN Seplos-BMS	Inter-cycle gap P95 (us)	165776.235	165777.835	168229.680	+1.600 (+0.0%)	+2453.445 (+1.5%)	+2451.845 (+1.5%)

3 Capture Overview

3.1 RS485/UART Capture Overview

Target	Topology		Frames	Complete	Incomplete	Req->Rsp avg (us)	Req->Rsp P95 (us)	Full P95 (us)
china tower modbus	Bridge		79	39	1	5015.035	5337.927	53525.724
daly rs485	Bridge		88	44	0	14458.976	18060.297	297646.802
growatt rs485	Bridge		50	25	0	4759.997	5294.317	53520.070
growatt seplos rs485	Bridge		49	24	1	49517.093	49863.536	97806.109
direct growatt rs485	Direct cable		100	50	0	5126.758	5438.647	75000.908
forward growatt rs485	Bridge	For-	50	25	0	5008.099	5293.782	53484.774
	ward							
forward growatt seplos rs485	Bridge	For-	49	24	1	49604.182	50490.317	97846.399
	ward							
jkbms modbus	Bridge		78	39	0	4903.504	5191.553	13115.590
pace modbus	Bridge		79	39	1	4987.736	5260.878	66064.784
pylon rs485	Bridge		49	18	5	4817.499	5537.460	148667.742
anenji pylon rs485	Bridge		20	8	4	9706.670	9912.793	153034.215
direct anenji jkbms pylon rs485	Direct cable		61	20	21	5124.941	5692.958	145154.533
direct anenji seplos pylon rs485	Direct cable		50	20	10	9887.208	12043.433	151498.699
forward anenji pylon rs485	Bridge	For-	24	8	8	5488.619	6190.079	149320.428
	ward							
forward anenji seplos pylon rs485	Bridge	For-	20	8	4	9469.731	9877.579	153001.644
	ward							
voltronic modbus	Bridge		78	39	0	99796.750	101837.191	119962.102
wow modbus	Bridge		78	39	0	4799.126	5093.604	65808.754

3.2 CAN Capture Overview

Target	Topology	Frames	IDs	Cycles	Cycle min (us)	Cycle avg (us)	Cycle max (us)	Cycle P95 (us)	Errors
deye can	Bridge	100	8	9	599848.765	1011013.334	1599872.420	1599871.234	0
goodwe can	Bridge	100	8	4	300245.895	2399973.046	7100062.150	6275024.243	0
growatt can	Bridge	519	10	1	9980699.265	9980699.265	9980699.265	9980699.265	0
growatt sepos can	Bridge	130	12	10	251234.865	851373.724	1496223.985	1221318.481	0
direct growatt can	Direct cable	260	10	1	4980232.085	4980232.085	4980232.085	4980232.085	0
direct growatt sepos can	Direct cable	65	12	6	129932.965	687125.942	831769.225	831768.787	0
forward growatt can	Bridge For-ward	520	10	1	9980680.500	9980680.500	9980680.500	9980680.500	0
forward growatt sepos can	Bridge For-ward	129	12	11	85691.125	755942.000	871814.905	861952.157	0
jkbms can	Bridge	344	12	1	9907992.715	9907992.715	9907992.715	9907992.715	0
pylon can	Bridge	100	8	5	900275.935	1900011.264	2299947.265	2299945.203	0
sma can	Bridge	60	6	60	219.400	230.733	237.400	237.400	0
sofar can	Bridge	148	6	148	219.400	228.765	237.400	237.400	0
victron can	Bridge	250	19	1	9960725.835	9960725.835	9960725.835	9960725.835	0

Notes

- The LaTeX source is generated from committed CSV files under `analysis/out/`.
- Capture durations are not identical, so normalized rates are included where useful.
- CAN cycle duration can be dominated by capture length when only one cycle is detected.
- Compiled PDFs and LaTeX build artifacts should stay out of the repository.